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10 October 1975
SUPERSEDING
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MILITARY SPECIFICATION

TROUSERS, WET SUIT (CHLOROPRENE FOAM)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a trouser, wet suit, fabricated from closed cell chloroprene foam.

1.2 Classification. The trousers shall be of one type only and of the following sizes (see 6.2).

Schedule of sizes

30-31, 32-33, 34-35, 36-37, 38-39

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-F-106 - Fasteners, Slide, Interlocking
V-T-285 - Thread, Polyester
DDD-L-20 - Labels; For Clothing, Equipage and Tentage (General Use)
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Box, Fiberboard
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams and Stitchings

FSC 4220

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS
D-1056 - Tests for Sponge and Expanded Cellular Rubber Products

(Copies of ASTM publications may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the supplier (see 6.3). Variations from this specification may appear in the sample, in which case the specification shall govern.

3.2 First article approval. The requirement for first article will be as specified by the procuring activity (see 6.2).

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3.3 Materials.

3.3.1 Chloroprene foam. The chloroprene foam for the trousers shall be pigmented black, expanded chlorinated cellular rubber, 3/16 (± 1/32) inch (4.8 ± .8 mm) thick, having a closed cell structure conforming to ASTM designation SCE41 and other requirements set forth herein when tested as specified in 4.3.1.1. It shall have a minimum density of 10 and a maximum density of 20 pounds per cubic foot (minimum density of 157.5 and a maximum density of 315 grams per cubic centimeter), with a natural skin surface on one or two sides. The foam used for the cuff facing and crotch and side reinforcements shall be the same as specified for the trousers, except that the thickness shall be 1/8 (± 1/32) inch (3.2 ± .8 mm) thick.

3.3.2 Knitted stretch cloth. The stretch cloth shall be circular knit from 100 percent continuous filament stretch-type nylon yarn and conform to the physical requirements of Table I when tested as specified in 4.3.1.1.

Table I - Knitted stretch cloth

Knitted stitch	Yarn per inch (min.) (Yarn per cm. (min.)) Wales	Courses	Weight, oz. per sq. yd. (g. per sq. m.)	Elongation, percent Wales	Courses	Color
Jersey	20 (7.9)	30 (11.8)	3.0 ± 0.25 (100.8 ± 7)	140 min. 200 max.	200 min. 280 max.	Black

3.3.2.1 Colorfastness. The knitted cloth shall show fastness to laundering, crocking and perspiration equal to or better than the standard sample. When no standard sample is available, the finished material shall show a minimum of "good" fastness to laundering, perspiration and a Munsell Value for crocking not lower than 8.5. Testing shall be as specified in 4.3.1.1.

3.3.3 Lamination. The stretch cloth shall be laminated to one side of the 3/16 inch (4.8 mm) chloroprene foam by means of a suitable adhesive, or other process capable of meeting the physical requirements of the laminated foam material. If the foam has no side with a cut surface, the stretch cloth shall be laminated on any side. When foam has a cut surface on one side, the stretch cloth shall be laminated to the cut side. The adhesive used shall be non-toxic, non-irritant and free from objectionable odor. The fabric backed (laminated) side shall be smooth and free of wrinkles or puckers, and show no signs of blocking, tackiness or adhesive strike-through. The 1/8 inch (3.2 mm) thick foam shall not be laminated with the stretch cloth.

3.3.3.1 Laminated foam. The finished laminated foam material shall conform to Table II when tested as specified in 4.3.1.1.

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Table II - Laminated foam

Characteristic	Requirement
Compression set, % maximum	35
Compression deflection, psi (g/cm^2)	2.0 (min.) (139) 7.0 (max.) (486.5)
Accelerated aging shrinkage, % maximum in each direction	5.0
Water absorption, % maximum	15.0
Laundering resistance	No delamination
Adhesion, original, lbs. (kg.) min., per two inches (50.8 mm)	6.0 (2.7)
Adhesion after launderings, lbs. (kg.) min., per two inches (50.8 mm)	6.0 (2.7)
Adhesion after aging, lbs. (kg.) min., per two inches (50.8 mm)	6.0 (2.7)
Blocking	No blocking

3.3.4 Thread, polyester. The thread used for sewing the slide fastener shall conform to type II, class 1, 4 ply, letter size F of V-T-285. The color of the thread shall be Black AA, Cable No. 66043.

3.3.4.1 Colorfastness. The dyed thread shall show fastness to laundering, perspiration and light equal to or better than the standard sample, when tested as specified in V-T-285. When no standard sample is available, the dyed thread shall show a minimum of "good" fastness to laundering, perspiration and light when tested as specified in V-T-285.

3.3.5 Slide fastener. The slide fastener used for the leg cuff closure shall be 9 inches (228.6 mm) long for all sizes and shall be non-separating, type I, style 3, size M, short tab pull. The metal parts shall be of nickel silver and conform to the requirements of V-F-106.

3.3.5.1 Tape. The tape for the slide fastener shall be cotton or cotton warp and nylon filling, twill or plain weave, waterproof and mildew resistant treated as specified in V-F-106. The extension beyond the scoops shall be a minimum of 1-3/8 inches (34.9 mm). All pull tabs shall be provided with a thong made of nylon tape or webbing. The color of both the tape and thong shall be black.

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3.3.6 Adhesive cement. The cement used to seal the seams shall be a chloroprene compound and shall conform to the requirements herein when tested as specified in 4.3.1.1.

3.3.7 Marking. The marking shall be applied on each trouser with a yellow marking medium conforming to type IV, class 1 of DDD-L-20. Marking shall be applied to the inside back of trousers, centered on the left side, one inch (25.4 mm) from top edge of trousers. The marking shall be permanent and shall not run, bleed, or rub off when tested as specified in 4.3.1.1. The inscription shall read as follows:

TROUSERS, WET SUIT (CHLOROPRENE FOAM)
CONTRACT NO.: DSA-100-00-0-000 (Example)
STOCK NO.: 4220-00-000-0000 (Example)
SIZE: 34-35 (Example)
NAME OF SUPPLIER:

3.3.7.1 Special marking. A white paper label printed or stamped with black block letters 1/8 inch (3.2 mm) high for capital letters and 1/16 inch (1.6 mm) high for lower case letters, bearing the following information shall be placed inside each trouser inside the right leg opening:

To prolong the life of the trousers, observe the following instructions for care and handling:

1. To don trousers open zippers and, if required, dust inside with talc to facilitate putting them on.
2. Sit down and work material up each leg until smooth. Do not pull and tug on trousers or the foam may tear.
3. Stand up, double over top of trousers, work material up until trousers lay smooth.
4. To remove, turn trousers inside out.
5. After each use, wash with fresh water and apply silicone or petroleum jelly on the slide fasteners.
6. Allow to dry before storage. Do not fold or crease in storage.

3.4 Design. The trousers shall be made of laminated closed cell chloroprene foam material (see 3.3) and shall have a high waist with leg bottoms that have slide fasteners attached adjacent to inseams. Fasteners shall be cemented, stitched and faced with lightweight unsupported chloroprene foam (see Figure 1).

3.4.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

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3.5 Patterns. Standard patterns will be furnished by the Government (see 6.3). The standard patterns shall not be altered in any way and are to be used for cutting the supplier's working patterns. The working patterns shall be identical to the standard patterns.

3.5.1 Pattern parts. The component parts of the trousers shall be cut from the material specified in accordance with the following pattern parts:

Material	Nomenclature of pattern parts	Cut parts
Laminated foam	Front	2
	Back	2
Unsupported foam	Cuff facing	2
	Side seam reinforcement	2
	Crotch reinforcement	1

3.6 Construction.

3.6.1 Stitch, seam and stitching types. Stitch, seam and stitching types specified in Table III shall conform to FED-STD-751.

3.6.2 Thread breaks and ends of seams. The ends of stitching on the slide fastener shall be securely tacked or backtacked not less than 3/8 inch (9.5 mm). Thread breaks shall be secured by stitching back of the break not less than 1/2 inch (12.7 mm).

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table III.

3.6.4 Cemented seams. The cemented seams shall have a minimum breaking strength of 28 pounds per two inches (12.6 kg per 50.8 mm) and the cemented reinforcement pieces shall have a minimum adhesion of 5.0 pounds per inch when tested as specified in 4.3.1.1.

3.7 Manufacturing requirements. The trousers shall be manufactured in accordance with the operation requirements specified in Table III. The supplier is not required to follow the exact sequence of operations listed provided that the finished trousers are identical to that produced by following the sequence of operations as listed in Table III. The smooth natural skin side of the foam rubber shall be used as the face of the material and shall be on the outside.

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Table III - Construction of trousers

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting of laminated foam material The trousers shall be cut in accordance with patterns furnished which shall show size, shape and markings for proper assembly of all component parts.					
2.	Replacement of damaged components During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced with non-defective and properly matched components.					
3.	Part markings a. All parts of the basic material shall be marked or ticketed to insure proper assembly and size. b. The use of metal fastening devices or sew-on tickets is prohibited. c. The use of ink pad numbering machine, rubber stamp or pencil is allowed providing the numbering does not show on the outside of the trousers.					
4.	Preparing components for assembly a. Apply two coats of chloroprene cement to all areas to be abutted, reinforced or stitched, 24 hours prior to assembly. b. Apply two additional coats of chloroprene cement to all areas to be abutted, reinforced or stitched, after the 24 hour period. (No solvent is required on seams.)					

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Table III - Construction of trousers (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Preparing components for assembly (cont'd)					
	c. The abutted seams shall be overcoated 1/4 (± 1/16) inch (6.4 ± 1.6 mm) beyond seams on the outside and on the inside of the trousers.					
5.	Join the slide fastener to leg bottom opening					
	a. Cement the closed slide fastener tapes to the inside bottom of left and right leg openings as indicated by marks on the pattern. The stops at the top of the slide fasteners shall be approximately 1-1/2 inches (38.1 mm) from the leg opening.					
8	b. Care shall be exercised not to apply cement on scoops of slide fasteners.					
	c. Double stitch 3/16 to 1/4 inch (4.8 to 6.4 mm) gauge, the tapes of slide fasteners with the inner row of stitching approximately 3/8 inch (7.5 mm) from the chain edge and continued across the bottom of slide fastener.	301	SSaa-2	4-5	F/4	F/4
6.	Assembly instruction					
	a. Insure that all seams to be cemented are cleaned of foreign matter and are tacky prior to joining.					
	b. When joining seams, allow more time for curing on difficult or critical curves prior to the next step of joining.					

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Table III - Construction of trousers (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
7.	Assembly a. Cement (abut) inseams up to crotch. b. Cement the cuff facings over the slide fasteners approximately 1-1/2 inches (38.1 mm) from edge of scoops. The smooth or uncut surface of the facings shall face toward the body when worn and shall be lapped over the slide fasteners towards the front of the legs. c. Cement (abut) side seams. d. Cement (abut) front and back center seams. e. Cement unsupported foam material reinforcement pieces centered over top end of side seams of trousers. f. Cement unsupported foam material reinforcement piece centered over the intersecting seams of the crotch on the inside of the trousers. g. Slit (cut) the leg openings 7 (± 1/2) inches (177.8 ± 12.7 mm) centered on the slide fastener.					
8.	Marking Markings shall be applied as specified in 3.3.7.					
9.	Cleaning a. Trim all thread ends. b. Remove all loose threads. c. Remove all excessive cement without injury to the material.					

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3.8 Sizes and measurements. The sizes and measurements of the finished trousers shall conform to the measurements shown in Table IV. All measurements and tolerances are expressed in inches and millimeters. The measurements shall be taken as indicated below (see Figure 1).

Table IV - Measurements

Size	Waist (A)	Total length (B)	Width across seat (C)
30-31	29 (737)	42-1/2 (1080)	35 (889)
32-33	31 (787)	43-1/8 (1095)	37 (940)
34-35	33 (838)	43-1/4 (1099)	39 (991)
36-37	35 (889)	43-3/8 (1102)	41 (1041)
38-39	37 (940)	43-1/2 (1105)	43 (1092)
Tolerance	± 1/2 (12.7)	± 1/2 (12.7)	± 1/2 (12.7)

NOTE: The measurements shall be taken with the trousers laid flat and face up in a relaxed condition.

Waist (A) - Double the measurement taken straight across the front of trousers from folded edge to folded edge at a point on each side fold located 1/2 inch (12.7 mm) below top of trousers.

Total length (B) - Length shall be measured from top of trousers in a line perpendicular to the bottom of leg opening.

Width across seat (C) - Double the measurement taken across the seat approximately 2 (± 1/4) inches (50.8 ± 6.4 mm) above crotch.

3.9 Workmanship. The finished trousers shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. When required, the first article submitted in accordance with 3.2 shall be inspected as specified in 4.3.2, for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Testing of components. In addition to the quality assurance provisions of the referenced specifications, components listed in Table VI shall be tested for the characteristics shown therein. The tests shall be performed on specially prepared cemented seams, similar to those made on the end item. The methods of testing specified in FED-STD-191 wherever applicable and as specified herein shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be in accordance with Table V and the lot shall be unacceptable if one or more units fail to meet any requirements specified. The basis for lot size for laminated and unlaminated chloroprene foam shall be one sheet, and one yard full width for the knitted stretch fabric. The sample unit for testing shall be as follows:

Unlaminated foam - one square yard (.8 square meter)
Laminated foam - one sheet
Knitted stretch fabric - one yard (.9 meter) full width

Table V - Sample size

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Table VI - Testing of components

Component	Characteristic	Requirement paragraph	Test method
Chloroprene foam	Material identification	3.3.1	1/
	Color	3.3.1	Visual 2/
	Thickness	3.3.1	ASTM D1056 3/
	Density	3.3.1	4.3.1.1.1
Knitted stretch cloth	Material identification	3.3.2	1/
	Filament	3.3.2	1/
	Weight	3.3.2	5040
	Wales and courses/inch (cm)	3.3.2	5070
	Elongation	3.3.2	4.3.1.1.2
	Color	3.3.2	Visual 2/
	Colorfastness:		
	Laundering	3.3.2.1	5614
	Crocking	3.3.2.1	5651
	Perspiration	3.3.2.1	5680
Laminated foam	Compression set	3.3.3.1	ASTM D1056 4/
	Compression deflection	3.3.3.1	ASTM D1056
	Shrinkage after accelerated aging	3.3.3.1	4.3.1.1.3
	Water absorption	3.3.3.1	4.3.1.1.4
	Laundering resistance	3.3.3.1	5556 6/
	Adhesion:		
	Initial	3.3.3.1	5950 5/
	After 3 launderings	3.3.3.1	5556 and 5950 5/ 6/
	After aging	3.3.3.1	4.3.1.1.3 and 5950 5/
	Blocking	3.3.3.1	5872 7/
Markings	Permanency	3.3.7	4.3.1.1.5
Adhesive cement	Material identification	3.3.6	1/
Cemented seam	Breaking strength	3.6.4	5100 8/

1/ A certificate of compliance will be accepted as evidence of conformance with this requirement.

2/ One determination shall be taken.

3/ Thickness - Except that 5 determinations shall be made.

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4/ Compression set - Test shall be run at room temperature for 22 hours with a recovery time of 24 hours.

5/ Adhesion - Except that the distance between the clamps shall be one inch (25.4 mm) at the start of the test. The use of a suitable solvent (e.g., MEK) to help separate the fabric from the foam sufficiently to fit the clamps is permissible. The adhesion shall be tested in the fabric's length (wale) direction only.

6/ Laundering - Except that five 8 by 8 inch (203 by 203 mm) specimens shall be tested and three cotton laundering cycles performed.

7/ Blocking - Except specimens 4 by 4 inches (102 by 102 mm) shall be placed together fabric side to fabric side.

8/ Cemented seams shall be centered equidistant from the clamps. All breaks shall be made with the seam at right angles to load direction. The clamps shall be one inch (25.4 mm) apart at the start of the test and the face of each jaw shall be not less than 1 by 2 inches (25.4 by 50.8 mm). The cemented seam test specimen shall be rectangular, 2 inches (50.8 mm) wide by not less than 6 inches (152.4 mm) long.

4.3.1.1.1 Density. Five specimens each 2 inches (50.8 mm) square shall be weighed separately to the nearest 0.001 gram on an analytical balance. The thickness of each specimen shall be determined in accordance with ASTM method D1056 and the following formula shall be used to determine density.

$$D = \frac{0.05 \times W}{T}$$

Where D = density in lbs/ft^3
W = the sum of the weights of the 5 specimens (grams)
T = the sum of the thicknesses of the 5 specimens (inches)

4.3.1.1.2 Elongation. An incline plane type of machine (i.e., I-P-4 or similar type) with a suitable autographic recording device shall be used. Specimens of the fabric shall be cut 4 by 6 inches (102 by 152 mm) with the long direction parallel to the direction to be tested. The specimen shall be evenly aligned without tension or slack in the 1 by 3 inch (25.4 by 76.2 mm) clamps set two inches (50.8 mm) apart. The machine shall run through two complete cycles, proceeding to its maximum depression so as to exert a five pound (2.25 kg) tension. Five specimens, each containing different wales and courses shall be tested, using individual recorded elongation graphs. The average stretch of five specimens for each direction shall be measured at the peak of the last recorded cycles, in a line perpendicular to the horizontal zero axis and reported to the nearest percent.

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4.3.1.1.3 Accelerated aging. Three square specimens each with a minimum dimension of 8 by 8 inches (203.2 by 203.2 mm) shall be measured to the nearest 1/32 inch (.8 mm) in both length and width direction by means of a steel rule. The specimens shall be suspended vertically in the air-oven at 158°F (± 2°F) (70°C ± 1°C) for seven days. At the conclusion of the aging interval, the specimens shall be cooled at room temperature on a flat surface and allowed to rest at least 4 hours before determination of shrinkage. The average shrinkage in each direction shall be calculated as follows and reported to the nearest 0.1 percent.

$$\text{Percent shrinkage} = \frac{O - A}{O} \times 100$$

Where O = original length or width (inches) (mm)
A = length or width after aging (inches) (mm)

4.3.1.1.4 Water absorption. Three specimens each 4 by 6 inches (102 by 152 mm) shall be weighed separately on an analytical balance to the nearest 0.1 gram. The specimens shall be placed in a tumble jar apparatus (as described in TM 5500 of FED-STD-191) containing two liters of distilled water and rotated for 20 minutes. At the end of the rotating period the specimens shall be immediately removed from the water, without pressure blotted dry of surface water with filter paper or other absorbent material and weighed. The average percent water absorption shall be calculated as follows and reported to the nearest 0.1 percent.

$$\text{Water absorption percent} = \frac{W2 - W1}{W1} \times 100$$

Where W1 is the weight of specimen before immersion
W2 is the weight of specimen after immersion

4.3.1.1.5 Marking test. The marking shall be tested by rubbing a wet, white cotton print cloth across the markings six times using normal pressure. There shall be no staining of the cotton cloth. The markings shall be complete in all details as specified in regard to correctness, completeness, legibility and location. Any of the above attributes that are not as specified shall be classified as a defect.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with 4.3.2.1 and 4.3.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.3.2.3. The sample unit shall be one finished trouser. The lot size shall be expressed in units of trousers.

4.3.2.1 Visual examination. The trouser shall be examined for defects in color, design, material, workmanship and marking and the defects classified in accordance with the list below.

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Defect	Classification Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
a. Any hole	X	
b. Any weakening defect such as deep creases or fold lines	X	
c. Needle chew	X	
d. Cut, tear or mend	X	
e. Unsightly cement mass (other than on fastener chain)		X
BASIC MATERIAL		
a. Any evidence of delamination	X	
b. Any brittle area which cracks when bent	X	
c. Area of excess cement, e.g., solid runs, ridges or lumps		X
d. Chloroprene foam or cement tacky	X	
e. Any imbedded foreign material easily removed resulting in damage of basic material	X	
f. Any imbedded foreign matter not easily removed, but if removable does not damage basic material		X
g. Any blister or pin hole	X	
h. Mottled, cloudy or streaky finish		X
i. Any objectionable odor		X
j. Reinforcement pieces omitted or misplaced	X	
CLEANNESS		
a. Excessive cement on chain or fastener	X	
b. One or more spots or stains of an obvious permanent nature		X
c. Thread ends not trimmed or loose thread ends not removed from trousers (to be scored only when the condition exists on major portion of trouser)		X

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Defect	Classification Major	Minor
COMPONENTS AND ASSEMBLY		
Any component part omitted or any required operation omitted or improperly performed (unless otherwise classified herein)	X	
CUTTING		
Any component part not cut in accordance with marks indicated on pattern or not in accordance with specification requirements	X	
SEAMS AND STITCHING		
a. Stitching omitted	X	
b. Not specified seam or stitch type		X
c. Less than or more than the required stitches per inch (mm)		X
d. Open seam	X	
e. Thread breaks not secured or stitching back of the break less than 1/2 inch (12.7 mm)		X
f. Ends of stitching on slide fastener not securely tacked or backstitched less than 3/8 inch (9.5 mm)		X
g. Part of garment caught in any unrelated stitching operation	X	
CEMENTED SEAMS		
a. Any open seam	X	
b. Uneven seam		X
c. Twisted or puckered seam	X	
d. Seam not properly joined	X	
MARKING AND SPECIAL MARKING LABEL		
a. Omitted, incorrect or illegible	X	
b. Color not as specified		X
c. Marking not permanent		X

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Defect	Classification Major	Minor
SLIDE FASTENER		
a. Not specified type, size, color or metal finish	X	
b. Other than cotton or cotton/nylon tape	X	
c. Tape stitched too loose or too tight, causing a noticeable bulge on tape or opening, when closed	X	
d. Stitching too close to chain edge not permitting sliders to pass	X	
e. Ends of chain uneven by:		
1. 1/4 inch (6.4 mm) or more but not more than 1/2 inch (12.7 mm)		X
2. More than 1/2 inch (12.7 mm)	X	
f. Chain length of slide fasteners less than 9 inches (228.6 mm) or more than 9-1/8 inches (231.8 mm)	X	
g. Tabs not provided with thongs		X
h. Thongs not made of nylon tape or webbing		X
NOTE: Slide fasteners shall be checked for proper function by opening and closing slide fastener at least three times.		
LEG CLOSURE SLIDE FASTENER		
a. Slide fastener on one or both leg openings omitted	X	
b. Closed slide fastener twisted, puckered or bulging	X	
c. Slide fastener stitched to leg with one row of stitching	X	
d. Leg cuff facings omitted	X	
e. Leg cuff facings tight or twisted causing fullness on leg opening	X	
f. Leg cuff facings not lapped over slide fastener as specified		X
g. Cement missing	X	
h. Leg opening not slit over center of fastener		X

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Defect	Classification Major	Minor
LEG CLOSURE SLIDE FASTENER (cont'd)		
i. Leg opening slit less than 6-1/2 inches (165.1 mm)		X
j. Leg opening slit more than 7-1/2 inches (190.5 mm)	X	

4.3.2.2 Dimensional examination. The trouser shall be examined for defects in dimensions in accordance with the following:

a. Any measurement deviating from nominal dimensions and tolerances specified shall be scored as a size measurement defect.

b. Leg lengths unequal in length by 1/2 inch (12.7 mm) or more shall be scored as a size measurement defect.

4.3.2.3 Inspection levels and acceptable quality levels. The inspection levels and acceptable quality levels (AQL's) expressed as defects per 100 units, for visual and dimensional examinations shall be as follows:

	AQL's	Inspection level
For examination in 4.3.2.1:		
Major	2.5	II
Total (Major and Minor combined)	6.5	II
For examination in 4.3.2.2:		
One class	6.5	S-3

4.3.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking comply with the Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping or inadequate stapling, bulging or distortion of container.
Content	Number of trousers per shipping container is more or less than required. 1/ Size shown on one or more trousers not as specified on packages or outer containers. 1/

1/ For this defect, two shipping containers from the sample shall be examined.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C (see 6.2).

5.1.1 Level A. Each pair of trousers shall be folded as follows: Close slide fasteners and lay garment flat, face up. Fold trousers in half by bringing the leg cuffs to the waistline. The folded trousers shall measure approximately 23 inches (584.2 mm) in length by 20 inches (508 mm) in width. To prevent the trouser legs from creasing, place a wadding of cellulosic cushioning material across the folded area of each trouser leg. The wadding shall be placed so that it bends easily inside the folded area. The wadding material shall be approximately 8 inches by 8 inches by 1 inch (203.2 mm by 203.2 mm by 25.4 mm) in thickness. Each folded pair of trousers shall be boxed in a fiberboard box conforming to style FTC, type CF, variety SW, class weather-resistant of PPP-B-636. The staples on the inside of the box body shall be covered with pressure sensitive tape to prevent abrading the trouser. Each folded pair of trousers shall be individually packaged in a fiberboard box approximately 24 inches (609.6 mm) in length, 21 inches (533.4 mm) in width, and 3 inches (76.2 mm) in depth. Approximate dimensions are furnished as a guide only. The box closure shall be secured with minimum 3 inch (76.2 mm) width gummed paper tape conforming to PPP-T-76 applied at the center of the length opening and extending across the bottom and up the side at least 3 inches (76.2 mm).

5.1.2 Level C (Commercial Packaging). Trousers shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Six pairs of trousers, of one size only packaged as specified in 5.1, shall be packed together in a close fitting box conforming to PPP-B-601, Overseas Type, Grade B, style optional. Strapping shall be zinc coated.

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5.2.2 Level B. Six pairs of trousers of one size only packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class weather-resistant, style RSC, grade V2s of PPP-B-636. Level A packages shall be packed flat one in length, one in width and six in depth within the shipping container. Inside dimensions of each shipping container shall approximate (guide only) 25 inches (635 mm) in length, 22 inches (558.8 mm) in width and 18-1/2 (469.9 mm) in depth. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636.

5.2.3 Level C (Commercial Packing). Trousers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The foam trouser is intended to be worn with the jacket, hood, mitten and bootee components of a wet-suit which is designed to protect personnel against the effect of cold water immersion.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Sizes required (see 1.2).
c. Whether first article sample is required (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

Custodian:
Navy - SA

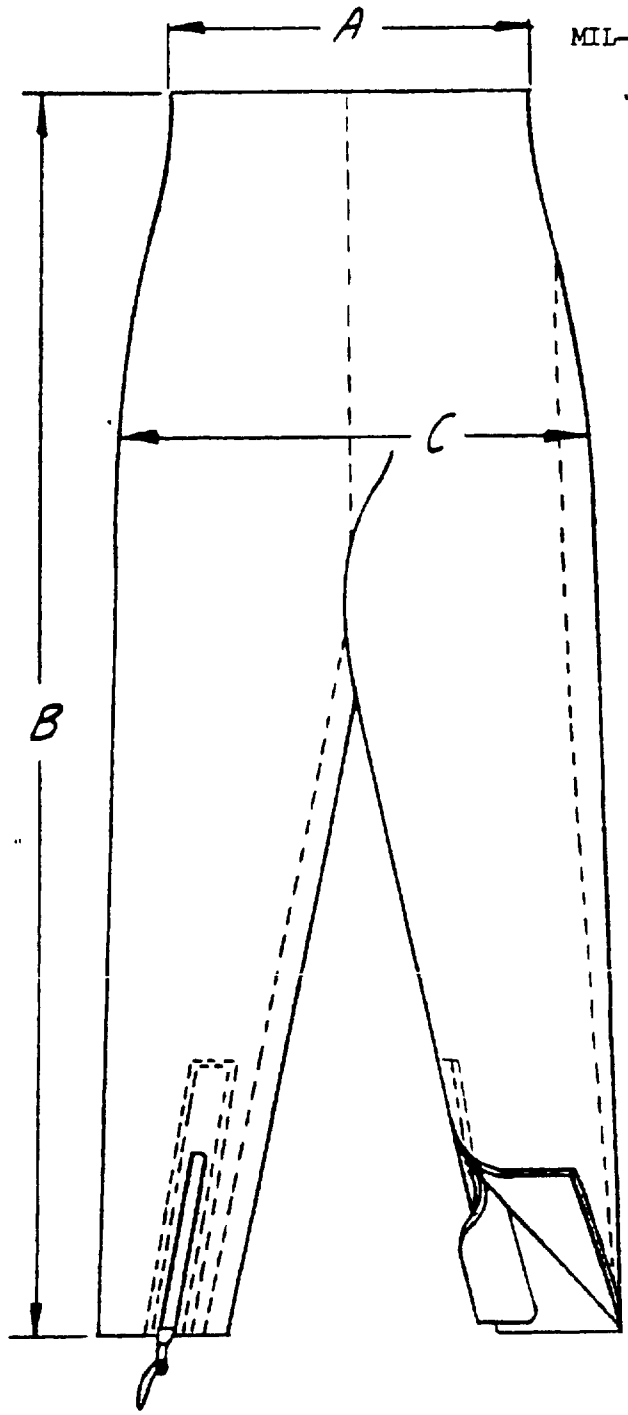
Preparing activity:
Navy - SA

Review activities:
Navy - MS
DSA - CS

Project No. 8415-0872

User activities:
Navy - AS, OS

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BACK VIEW

FIGURE 1 TROUSERS, WET SUIT, (CHLOROPRENE FOAM)

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FOLD

Officer in Charge
Navy Clothing and Textile Research Unit
21 Strathmore Road
Natick, MA 01760

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Officer in Charge
Navy Clothing and Textile Research Unit
21 Strathmore Road
Natick, MA 01760

FOLD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-82270B(SA); Trousers, Wet Suit (Chloroprene Foam)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

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